BOARD OF SECONDARY EDUCATION, KARACHI

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2024

CHEMISTRY

Past Paper 2024

Retyped faithfully from the original board paper — same questions, no changes.

SECTION "A" MULTIPLE CHOICE QUESTIONS (MCQs)

Maximum Marks: 12

Note: Note: (i) Attempt all the Questions of this Section. (ii) Do not copy down the part questions. Write only the answer against the proper number of the Question and its part according to the question Paper. (iii) Each question carries 1 Mark.

Q1. 1. Choose the correct answer for each question from the given options:

Q1. Natrium is the Latin name of this element:

(A) Nitrogen
(B) Neon
(C) Sodium
(D) Nickle

Q2. In chemical reaction ZnO + H2 → Zn + H2O, it is reduced: : (Reduced) ZnO + H2 → Zn + H2O

(A) H
(B) H2
(C) H2
(D) H

Q3. CO3–2 is a/an: : CO3–2

(A) cation
(B) anion
(C) molecule
(D) molecular ion

Q4. Number of sub-shells in M-shell is: : M -

(A) one
(B) two
(C) three
(D) four

Q5. Boron is a:

(A) metalloid
(B) metal
(C) non-metal
(D) transition metal

Q6. Atomic radius is measured in: : ( )

(A) yard
(B) feet
(C) inch
(D) angstrom

Q7. Shielding effect from left to right in a period:

(A) increases
(B) decreases
(C) same
(D) moderate

Q8. This is non-polar covalent compound:

(A) HF
(B) H2O
(C) H2O
(D) HF

Q9. At room temperature this metal exists in liquid state:

(A) Bromine
(B) Mercury
(C) Iron
(D) Calcium

Q10. This term would be the same for 1 mole of oxygen gas and 1 : mole of water:

(A) atoms
(B) mass
(C) molecules
(D) volume

Q11. Alkaline earth metals belong to this group:

(A) I–A
(B) II – A
(C) II – A
(D) I – A

Q12. This gas turns lime water milky:

(A) O2
(B) NO2
(C) NO2
(D) O2

SECTION “B” (Short Answer Question)

Maximum Marks: 24

Note: Note: Answer any EIGHT questions from this section. Each question carries 3 marks.

Q2. Define the following terms: (iii) (ii) (i) :

(i) Periodicity

(ii) Electronegativity

(iii) Electrolysis

Q3. Write down three differences between Alkali metals and Alkaline earth metals.

Q4. The following statements belong to which branch of Chemistry? Identify: (i) That chlorofluoro compounds are responsible for depletion of Ozone layer. (ii) That investigates the properties of (i) Ammonia gas. (iii) That find out the quantity and quality of (ii)

(iii) Paracetamol in Panadol tablet.

Q5. How do you preparer 100ml of 0.4M MgSO4 from a stock solution 0.2M MgSO4 0.4M MgSO4 100ml of 2.0M MgSO4?

Q6. What is atomic radius? Discuss its trend across the period and group in the periodic table.

Q7. What are isotope? Describe briefly three isotopes of Uranium.

Q8. Balance the following chemical equations by inspection method:

(i) CaCO3 + HCl → CaCl2 + H2O + CO2 i) CaCO3 + HCl → CaCl2 + H2O + CO2)

(ii) KClO3 → KCl + O2

(iii) NH3 + O2 → NO + H2O ii) KClO3 → KCl + O2)

Q9. Calculate the number of atoms presents in 9.2 g of Calcium.) 9.2 (Atomic mass of Ca = 40) (40 =

Q10. Draw electronic dot and cross structures or following molecules: (i) (i) H2(ii) : H2 (ii) N2 (iii) O2 (Atomic number of H = 1, N = 7, O = 8) (H=1 N=7 O=8 N) 2 (iii) O2

Q11. Write down the chemical formulae of the following compounds:

(i) Caustic Soda

(ii) Sand

(iii) Sugar (iii) (ii) (i)

(iv) Lime Stone

(v) Gypsum

(vi) Baking Soda (vi) (v) (iv)

Q12. What is meant by Tyndall effect? How is it be used for the identification of a true solution and a colloidal?

Q13. Justify the following:

(i) Why Fluoride ion is not considered as Neon atom? (i)

(ii) Why noble gases do not react with other elements at ordinary conditions. (ii)

SECTION “C” : (Detailed –Answer Question)

Maximum Marks: 24

> Note: Note: Attempt any FOUR questions from this section. Each question carries 6 marks.

Q14. State Charles law. A 600 ml sample of gas is heated from 27°C to C°77 C°27 600ml 77°C at constant pressure. What would be its final volume?

Q15. Define electroplating. Describe the Tin plating of an Iron spoon with the help of ionic equations. Draw its labelled diagram also.

Q16. Identify the highly electronegative group of Periodic table. Enlist the elements present in this group. Give three common characteristics of these elements.

Q17. Write down six postulates of Bohr's Atomic Model.

Q18. What is an ionic bond? Explain the formation of ionic bond between Sodium and Chlorine ionic compound Sodium chloride is formed.

Q19. What are molecules? Discuss its five types with examples.